Assignment 01

Random Experiments & Sample Spaces

Name: ..... ID: .....

Group: ..... Date: .....

> **Before you start.** This sheet stops before probability. Nothing here asks for a fraction — every answer is a **set** and a **count**. Two lines per question: list S in braces, then write $N = |S|$. For a multi-stage experiment, say in one phrase whether the stages are **identical** (multiply) or **different by branch** (add), as in Example 7 of the lecture.

Part A — Select the correct answer

1) Circle one option in each of the following.

i. The sample space of tossing a coin twice has how many elements?

A) 2 **B)** 3
C) 4 **D)** 8

ii. Which of the following is **not** an outcome of rolling a die once?

A) 1 **B)** 4
C) 6 **D)** 7

iii. A sample space that contains an interval of real numbers is called

A) discrete **B)** continuous
C) finite **D)** empty

iv. A coin is tossed and a die is rolled. Then $N =$

A) 8 **B)** 12
C) 36 **D)** 2

v. The sample space of the 2-digit numbers formed from the digits $\{2, 3\}$ is

A) $\{23, 32\}$ **B)** $\{22, 33\}$
C) $\{22, 23, 32, 33\}$ **D)** $\{2, 3\}$

Part B — Single-stage experiments

For each experiment, write the sample space S in full and state $N = |S|$.

2) A coin is tossed once.

3) A die is rolled once.

4) A ball is selected from a box containing 3 symmetric balls — red, yellow and green.

5) A 2-digit number is formed using the digits $\{2, 3\}$, repetition allowed.

6) A letter is selected at random from the word **PROBABILITY** and the letter observed is recorded.

Part C — Multi-stage experiments

7) Two dice are rolled. State N, and list every outcome in which the first die shows a 2.

8) A 2-digit number is formed from the digits $\{1, 2, 3\}$. Write S and N when
   i. repetition is allowed;
   ii. no digit may be repeated.

9) A coin is flipped. If a head occurs the coin is flipped a second time; if a tail occurs a die is rolled once. Write S and count it, and say why the multiplication rule does not apply.

10) A box contains one red ball and one blue ball. Two balls are drawn in order. Write S and N when
   i. the first ball is replaced before the second is drawn;
   ii. the first ball is not replaced.

11) A student writes the sample space of tossing a coin twice as $S = \{HH, HT, TT\}$ with $N = 3$. Explain in one sentence what is wrong, and draw the tree that settles it.

Part D — Discrete or continuous?

12) Classify each sample space as **discrete** or **continuous**, with a one-line justification.
   i. The suit of a card drawn from a deck.
   ii. The time in minutes a customer waits in a queue.
   iii. The number of cars passing a checkpoint in one hour.
   iv. The mass in kilograms of a randomly chosen apple.
   v. The number of tosses of a coin up to and including the first head.
   vi. The temperature recorded by a sensor, known to lie between $0°$ and $100°$.

Part E — Exercise set 1

13) Write the sample space for tossing a coin three times. How many outcomes?

14) A coin is tossed and a die is rolled. Write S, and say why it has 12 elements.

15) An experiment rolls a die; if the result is 6 the die is rolled again, otherwise a coin is tossed. Write S and count it.

16) Is the sample space of "the time in seconds until the next bus arrives" discrete or continuous? Justify in one sentence.